TERRAPROOF | SATELLITE EVIDENCE DOSSIER

Autonomous Geospatial Audit & Decision-Support System • Generated 2026-10-30 20:45:04 UTC

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Audit Dossier ID:	test-job-uuid-12345	**Analysis Type:**	INSURANCE VERIFICATION
Claim Reference:	TEST-REF-9921	**State / District:**	Punjab / Ludhiana
Farmer / Insured:	Sunita Devi	**Crop & Season:**	Soybean (Kharif)
Khasra / Land Record:	N/A	**Sum Insured / ha:**	■35,000/ha
Recommended Verdict:	**MANUAL_FIELD_INSPECTION_RECOMMENDED**	**Evaluation Standard:**	PMFBY Geospatial Audit Standard

1. EXECUTIVE EVIDENCE FINDINGS

Satellite analysis observed approximately 440.236 ha (100.0% of field) exhibiting spectral signatures consistent with flood.

Metric	Value	Methodological Meaning
Baseline Mean Index	0.8839	Mean radiometric canopy baseline prior to reported event
Post-Event Mean Index	-0.3386	Mean radiometric response observed immediately following event
Index Difference (Delta)	1.2225	Magnitude of spectral divergence (positive = loss/inundation)
Estimated Affected Area	**440.236 ha**	Total hectares showing thresholded anomalous change
Claimed Loss Area	8.4 ha	Hectares reported in submission form
Confidence Score	**88%**	High radiometric contrast observed between clear pre-event and post-event Sentinel-2 acquisitions. Corroborated by local meteorological precipitation trends.

2. METEOROLOGICAL CONTEXT (OPEN-METEO ARCHIVE)

Ground coordinates (27.11, 76.21) recorded **68.4 mm** total rainfall during the target window, peaking at 41.2 mm/day. Average topsoil moisture was estimated at 0.38 m³/m³.

3. SATELLITE SENSORS & ACQUISITIONS USED

Scene ID	Sensor	Acquisition Date	Cloud Cover
DEMO_S2A_MSIL2A_20260710_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-10T10:30:00Z	4.2%
DEMO_S2B_MSIL2A_20260710_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-10T10:35:00Z	8.5%
DEMO_S2A_MSIL2A_20260715_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-15T10:30:00Z	4.2%
DEMO_S2B_MSIL2A_20260715_T43RE...	Sentinel-2 MSI (Demonstration Fixture)	2026-07-15T10:35:00Z	8.5%

4. POTENTIAL INCONSISTENCIES & AUDIT FLAGS

• ***AUDIT FLAG:*** *Observed affected area (440.236 ha) diverges by 5140.9% from reported claimed area (8.4 ha).*

5. METHODOLOGICAL LIMITATIONS & RESPONSIBLE USE

• Satellite pixel resolution is 10 meters; sub-pixel boundary mix may introduce +/- 8% area uncertainty.
• Optical observations can be occluded by cloud cover; review cloud metadata flags for each acquisition date.
• Surface greenness (NDVI) can decrease due to regular crop harvest or seasonal senescense rather than disaster.
• Weather data from nearby grid cells may differ from localized cloudburst or field microclimate.

6. ESTIMATED COMPENSATION SCENARIO (PMFBY MODEL)

Sum Insured / ha	■35,000	**Assumed Severity %**	80.0%
Observed Affected Area	440.24 ha	**Estimated Scenario Amount**	**■12,326,608**

Formula: ■35,000/ha × 440.24 ha × 0.80 = ■12,326,608 • Illustrative scenario estimate for decision support only. Actual insurance settlement depends on applicable PMFBY scheme guidelines, notified yield thresholds, and survey committee approval.

7. HUMAN ADJUSTER AUDIT & DECISION SIGN-OFF

Adjuster Decision:	PENDING ACCREDITED REVIEW	**Date of Inspection:**	_____
Adjuster Name & ID:	_____ __	**Physical Signature:**	_____

LEGAL & FORENSIC DISCLAIMER: *TerraProof is a technical decision-support and evidence-gathering instrument. Satellite spectral indices (such as NDVI, NDWI) provide indirect radiometric indicators of vegetation and surface moisture. This dossier does NOT constitute a legally binding verdict of fraud or guarantee of compensation. Final claim approval or denial remains the exclusive responsibility of accredited human insurance adjusters.*